Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 21)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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5	
6	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 A student measured the mass of carbon stored per hectare in three land uses in the same region.

Table 1

Land use	Carbon in vegetation / t ha^{-1}	Carbon in soil dead organic matter / t ha^{-1}
Mature woodland	120	95
Permanent pasture	3	80
Arable (ploughed annually)	2	45

0 1 . 1 Calculate the total carbon lost per hectare when mature woodland is converted to arable farmland.

[1 mark]

Answer ☐ t/ha

0 1 . 2 Explain why the soil carbon in arable land is so much lower than in pasture, even though the vegetation carbon is similar.

[2 marks]

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0 1 . 3 Suggest **two** ways the arable farmer could increase soil carbon storage.

[2 marks]

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2 .....

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0 2 . 1 Explain how each of the following features of vehicle design reduces energy use.

[6 marks]

Aerodynamics .....

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Reduced mass .....

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Regenerative braking (KERS) .....

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0 2 . 2 Explain why solid steel wheels on rails lose less energy than pneumatic tyres on roads.

[2 marks]

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0 2 . 3 Describe **two** ways combustion efficiency in an internal combustion engine can be improved.

[2 marks]

1 .....

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2 .....

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0 3 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 3 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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0 4 Table 1 shows the effect of installing water meters on the mean daily water use of households in a UK town.

Table 1

	Unmetered households (fixed charge)	**Metered households**
Mean water use / dm^3 per person per day	150	125
Percentage with low-flush toilets	35	58
Percentage using a garden hosepipe weekly in summer	44	20

0 4 . 1 Calculate the percentage reduction in water use per person in metered households.

[1 mark]

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Answer [ ] %

0 4 . 2 Explain why metering reduces water use.

[2 marks]

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0 4 . 3 Describe **three** ways in which domestic water use can be reduced other than metering.

[3 marks]

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0 4 . 4 Explain why about 20% of the water intended for public supply in the UK never reaches consumers, and how this could be reduced.

[2 marks]

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0 4 . 5 Explain how drip irrigation reduces the water needed to grow a crop compared with spray irrigation.

[2 marks]

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0 5 A mining company is assessing a copper deposit beneath farmland.

0 5 . 1 Describe what each of the following survey techniques detects.

[3 marks]

Gravimetry .....

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Magnetometry .....

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Seismic surveys .....

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0 5 . 2 Explain why trial drilling is normally carried out only after the remote sensing surveys.

[2 marks]

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0 5 . 3 Explain how a rise in the world price of copper can turn part of the **resource base** into a **reserve**.

[2 marks]

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0 5 . 4 Explain **four** environmental impacts of extracting the copper by opencast mining.

[4 marks]

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0 5 . 5 Describe how the site could be restored after closure, and explain why restoration is planned before mining begins.

[4 marks]

15

0 6 . 1 Describe the two processes that drive the North Atlantic Conveyor.

[2 marks]

0 6 . 2 Explain how global climate change could weaken the North Atlantic Conveyor and describe the possible consequence for the climate of the UK.

[4 marks]

Do not write outside the box

0 6 . 3 Explain why the effects of climate change on UK temperatures are difficult to predict.

[3 marks]

0 6 . 4 Describe how deep ocean currents are monitored.

[1 mark]

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0 7 A student used 0.5 m x 0.5 m grid quadrats to survey ground flora in two meadows, one grazed and one ungrazed. Table 1 shows the results for the grazed meadow (10 quadrats).

Table 1

Species	Number of individuals
Ribwort plantain	48
Perennial ryegrass	120
White clover	36
Daisy	24
Creeping buttercup	12

0 7 . 1 Explain why a 0.5 m x 0.5 m quadrat was chosen rather than a 10 cm x 10 cm or 100 m x 100 m quadrat.

[2 marks]

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0 7 . 2 Calculate Simpson's Index of Biodiversity (D) for the grazed meadow using D = N(N-1) / Σn(n-1). Show your working.

[3 marks]

Answer

0 7 . 3 The ungrazed meadow had D = 1.8 with the same number of species. Explain what this shows.

[2 marks]

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0 7 . 4 Explain why a point quadrat would be better than a frame quadrat in tall vegetation.

[2 marks]

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0 7 . 5 Give **one** limitation of using the DAFOR abundance scale.

[1 mark]

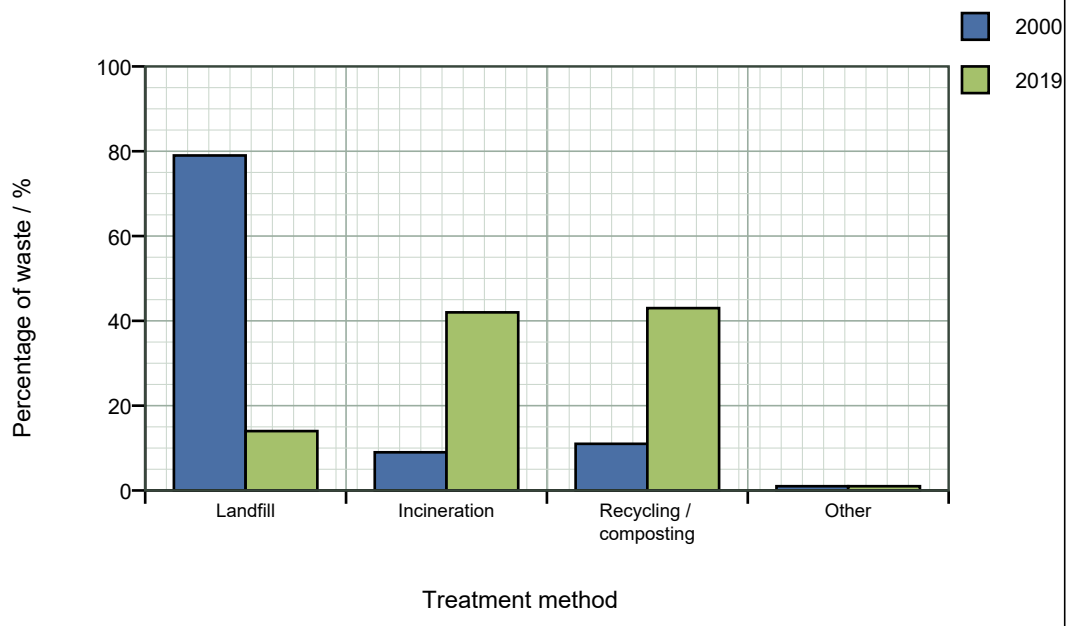
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0 8 Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1



0 8 . 1 Describe the changes shown in Figure 1.

[2 marks]

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0 8 . 2 Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

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0 8 . 3 Describe **four** features of a well-managed landfill site.

[4 marks]

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0 8 . 4 Suggest why incineration has increased more than recycling for some waste types.

[1 mark]

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0 9 . 1 Which chemical form of mercury is most easily absorbed and biomagnified in food chains? Tick **one** box.

[1 mark]

A Elemental (liquid) mercury ☐

B Inorganic mercury compounds ☐

C Organic (methyl) mercury ☐

D Mercury vapour ☐

0 9 . 2 Explain why anaerobic sediments increased the hazard of the inorganic mercury discharged at Minamata.

[2 marks]

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0 9 . 3 Explain why heavy metal wastes are stored under alkaline conditions.

[2 marks]

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1 0 Table 1 shows the mean daily solar energy received on a horizontal surface at three locations.

Table 1

Location	Latitude	Mean solar energy in June / kWh m^{-2} day^{-1}	Mean solar energy in December / kWh m^{-2} day^{-1}
Sahara desert	25°N	7.8	4.2
Southern England	51°N	5.0	0.7
Northern Norway	69°N	4.6	0.0

1 0 . 1 Use Table 1 to calculate how many times more solar energy is received in December in the Sahara than in southern England.

[1 mark]

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Answer [ ] times

1 0 . 2 Explain why northern Norway receives more solar energy in June than in December.

[2 marks]

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1 0 . 3 Explain why concentrating solar power systems using parabolic reflectors only work where there is no cloud.

[2 marks]

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1 0 . 4 Explain why the low energy density of solar power is a problem for large-scale generation.

[1 mark]

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1 0 . 5 Evaluate the extent to which new technologies could overcome the problems of using solar power as a large-scale energy resource.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS